

Jessie Yuan

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Education

Carnegie Mellon University (CMU) – Pittsburgh, PA

Aug. 2022 – May 2026

B.S. in Computer Science, Minor in Mathematics

- GPA: 4.0/4.0
- **Relevant Coursework:** Convex Optimization (10-725)*, Robot Learning (16-831), Fundamentals of Human-Robot Interaction (16-701), Machine Learning (10-701), Embodied AI Safety (16-886), Algorithm Design and Analysis (15-451)

Research Experience

Interactive and Trustworthy Robotics (Intent) Lab – Robotics Institute, CMU

May 2025 – Present

Undergraduate Researcher, Advised by Prof. Andrea Bajcsy

- Implemented system using world models and VLMs to translate rollouts from a diffusion policy into textual narrations and select appropriate robot behaviors based on nuanced language instructions.
- Applied conformal prediction to quantify VLM uncertainty, distinguishing between scenarios where multiple valid behaviors exist versus when the base policy cannot accomplish the task.
- Tested various methods to improve confidence score quality, ultimately implementing marginalization over user intent to explicitly model ambiguity in human commands.
- Developed a residual learning pipeline enabling user correction of base policy failures and iterative policy improvement from collected corrections.

Robotic Caregiving and Human Interaction (RCHI) Lab – Robotics Institute, CMU

Feb. 2023 – May 2025

Undergraduate Researcher, Advised by Prof. Zackory Erickson

- Developed a ChatGPT-driven speech interface for a commercial feeding robot through prompt engineering and awarded CMU Summer Undergraduate Research Fellowship for this project.
- Designed a ROS-based system to collect, label, and process data from wearable sensors; cleaned, visualized, and performed feature extraction on this data; and trained and deployed binary classification and regression models using PyTorch.
- Designed, organized, and conducted three human studies, including procedure documentation, participant interaction, and data analysis to collect training data and evaluate system efficacy and user experience.

Publications

J. Yuan*, Y. Wu, and A. Bajcsy, “When to Act, Ask, or Learn: Uncertainty-Aware Policy Steering”, *Under Review*, 2026.

A. Padmanabha*, **J. Yuan***, T. Mehta, R. K. Jenamani, E. Hu, V. de León, A. Wertz, J. Gupta, B. Dodson, Y. Yan, C. Majidi, T. Bhattacharjee[†], and Z. Erickson[†], “WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding”, accepted to *The ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2026.

A. Padmanabha*, **J. Yuan***, J. Gupta, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, “VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots”, in *The ACM Symposium on User Interface Software and Technology (UIST)*, 2024. **Oral Presentation.**

J. Yuan, J. Gupta, A. Padmanabha, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, “Towards an LLM-Based Speech Interface for Robot-Assisted Feeding”, in *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST Adjunct)*, 2024. **Live Demonstration.**

Teaching Experience

Teaching Assistant for 21-128: Mathematical Concepts and Proofs, CMU

Fall 2024

Teaching Assistant for 15-251: Theoretical Computer Science, CMU

Spring 2024, Fall 2023

